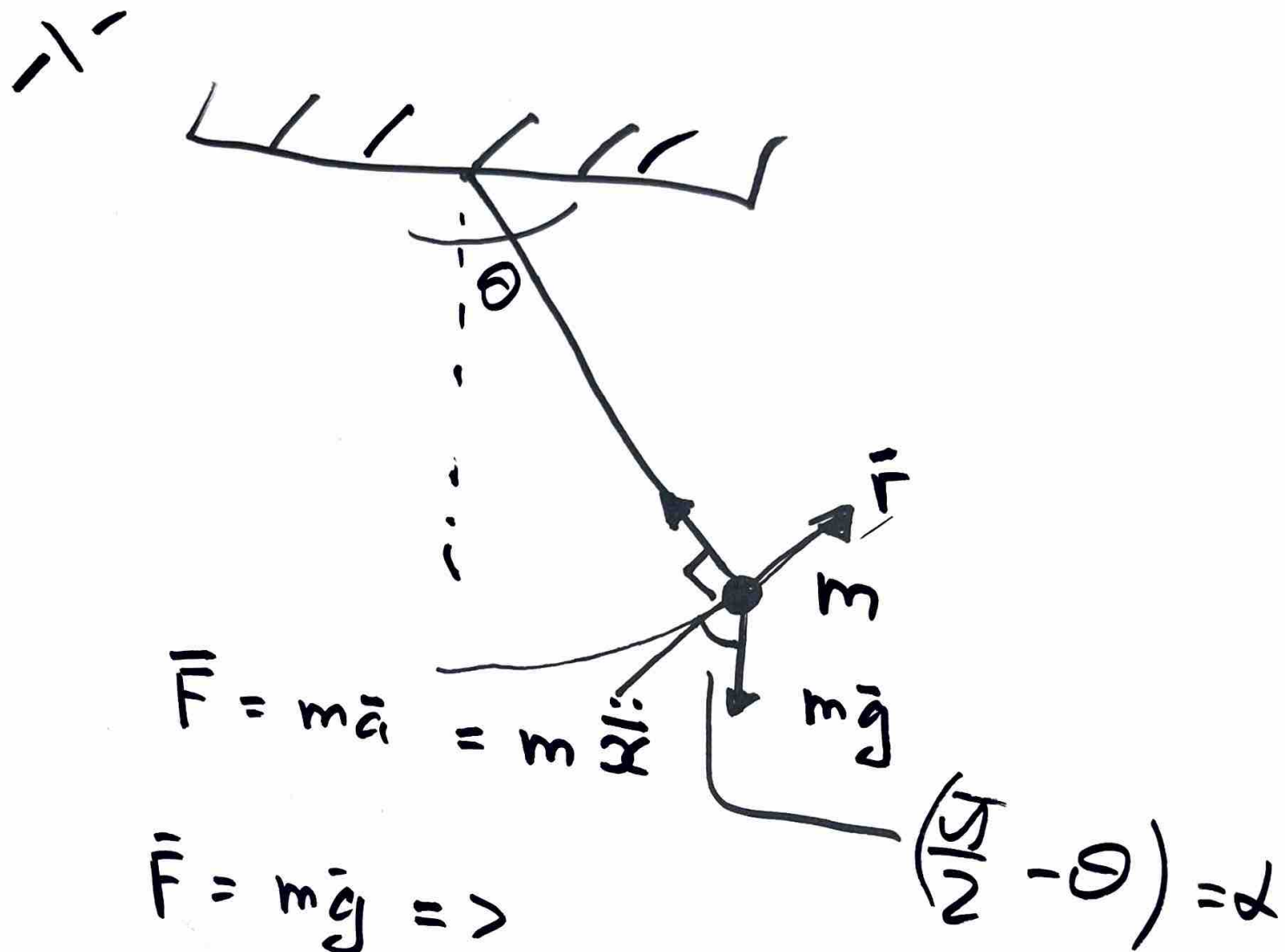




$$-mg \cos \alpha = -mg \sin \theta = ma = mL\ddot{\theta}$$

$$mL\ddot{\theta}(t) + mg \sin \theta(t) = 0$$

$$\ddot{\theta} + \frac{g}{L} \sin \theta = 0$$

$$\sin \theta \approx \theta$$

$$\ddot{\theta} + \frac{g}{L} \theta = 0$$

-2-

$$\ddot{\theta} + \frac{g}{L} \sin \theta = 0$$

$$t = T \tau$$

t, τ independent variables

$$[t] = \text{Time}$$

T is a const

$$[T] = \text{Time}$$

$$[\tau] = 1$$

$$\frac{d}{dt} = \frac{d\tau}{dt} \frac{d}{d\tau} = \frac{1}{T} \frac{d}{d\tau}$$

$$\frac{d^2}{dt^2} = \frac{1}{T^2} \frac{d^2}{d\tau^2}$$

$$T^2 = \frac{L}{g}$$

$$\frac{d^2}{d\tau^2} \theta(\tau) + \frac{g T^2}{L} \sin \theta(\tau) = 0$$

-3-

$$\left[\frac{d^2}{dT^2} \Theta(T) + \sin \Theta(T) = 0 \right] \frac{d\Theta}{dT}$$

$$\begin{cases} \frac{d}{dT} \Theta(T) = \omega(T) \\ \frac{d}{dT} \omega(T) = -\sin \Theta(T) \end{cases}$$

$$\ddot{\Theta} \dot{\Theta} + \dot{\Theta} \sin \Theta = 0$$

$$\left(\frac{d^2}{dT^2} \Theta(T) \right) \frac{d}{dT} \Theta(T) = \frac{1}{2} \frac{d}{dT} (\Theta(T))^2$$

$$\ddot{\Theta} \dot{\Theta} = \frac{d}{dT} \frac{\dot{\Theta}^2}{2}$$

$$\dot{\Theta} \sin \Theta = -\frac{d}{dT} \cos(\Theta)$$

$$-4- \quad \dot{\Theta} \ddot{\Theta} + \dot{\Theta} \sin \Theta = 0 \quad \rightarrow -\frac{d}{d\tau} \cos \Theta$$

$$L = \frac{d}{d\tau} \frac{\dot{\Theta}^2}{2}$$

$$\frac{d}{d\tau} \frac{\dot{\Theta}^2}{2} - \frac{d}{d\tau} \cos \Theta = 0$$

$$\frac{d}{d\tau} \left[\frac{\dot{\Theta}^2}{2} - \cos \Theta \right] = 0$$

$$\frac{\dot{\Theta}^2}{2} - \cos \Theta = E$$

$$\frac{d}{d\tau} \Theta(\tau) = \sqrt{2(E + \cos \Theta)}$$

-5-

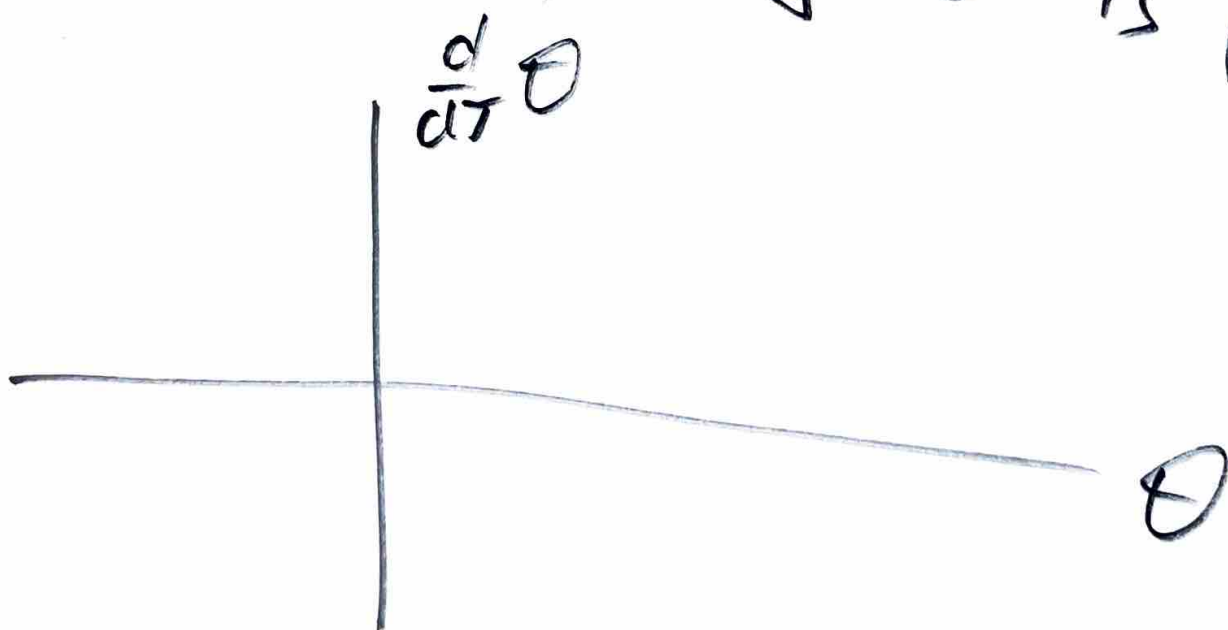
$$\frac{d}{dT} \Theta(T) = \sqrt{2(E + \cos \Theta)}$$

$$\text{IF } |E| < 1$$

$$\cos \Theta_{\max} = -E$$

$$\Theta_{\max} = \text{Arc Cos } E$$

IF $|E| > 1$, any Θ is possible



6 Flow on a LINE

$$\frac{d}{dt} x(t) = f(x(t)) - \text{autonomous}$$

Nonlinear, first order ODE

Most general form

$$\frac{d}{dt} x(t) = f(x(t), t)$$

non-autonomous

$$\dot{x}(t) = \sin x(t)$$

$$\frac{dx}{dt} = \sin x$$

$$\int \frac{dx}{\sin x} = \int dt$$

$$\int \frac{dx}{\sin x} = \ln \left(\tan \left(\frac{x}{2} \right) \right)$$

$$\frac{d}{dx} \left[\ln \left(\tan \left(\frac{x}{2} \right) \right) \right] = \frac{1}{2 \tan \left(\frac{x}{2} \right) \cos^2 \left(\frac{x}{2} \right)} =$$

$$= \frac{1}{2 \frac{\sin \left(\frac{x}{2} \right)}{\cos \left(\frac{x}{2} \right)} \cos^2 \left(\frac{x}{2} \right)}$$

$$= \frac{1}{2 \sin \left(\frac{x}{2} \right) \cos \left(\frac{x}{2} \right)} = \frac{1}{\sin x}$$

-8-

$$t+c = \ln(\tan(x/2))$$

$$e^{t+c} = \tan \frac{x}{2}$$

$$2 \operatorname{ArcTan}(e^{t+c}) = x(t)$$

$$T = \sqrt{\frac{L}{g}}$$

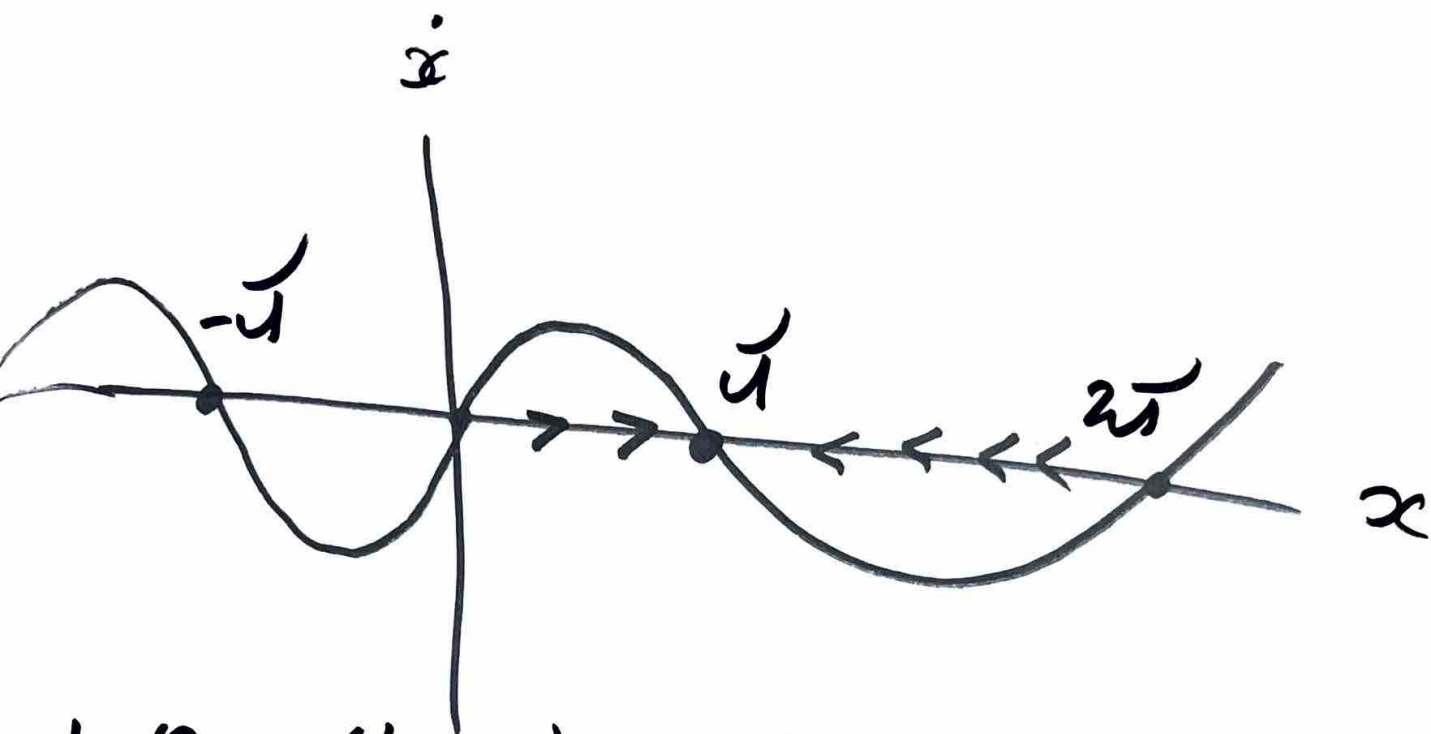
① Let $x(t=0) = \pi/2$
 $x(t) = ?$

② what happens with $x(t)$
as $t \rightarrow \infty$

③ Types of trajectories

-9-

$$\dot{x} = \sin x$$



• Let $0 < x(t=0) < \pi$

$$0 < x(t=0) < \pi$$

$$\dot{x}(t=0) > 0$$

$x(t)$ grows until $x = \pi$, stay at π

• Let $\pi < x(t=0) < 2\pi$

$$\dot{x}(t \neq 0) < 0$$

$x(t)$ will decrease until $x(t) = \pi$, stay there

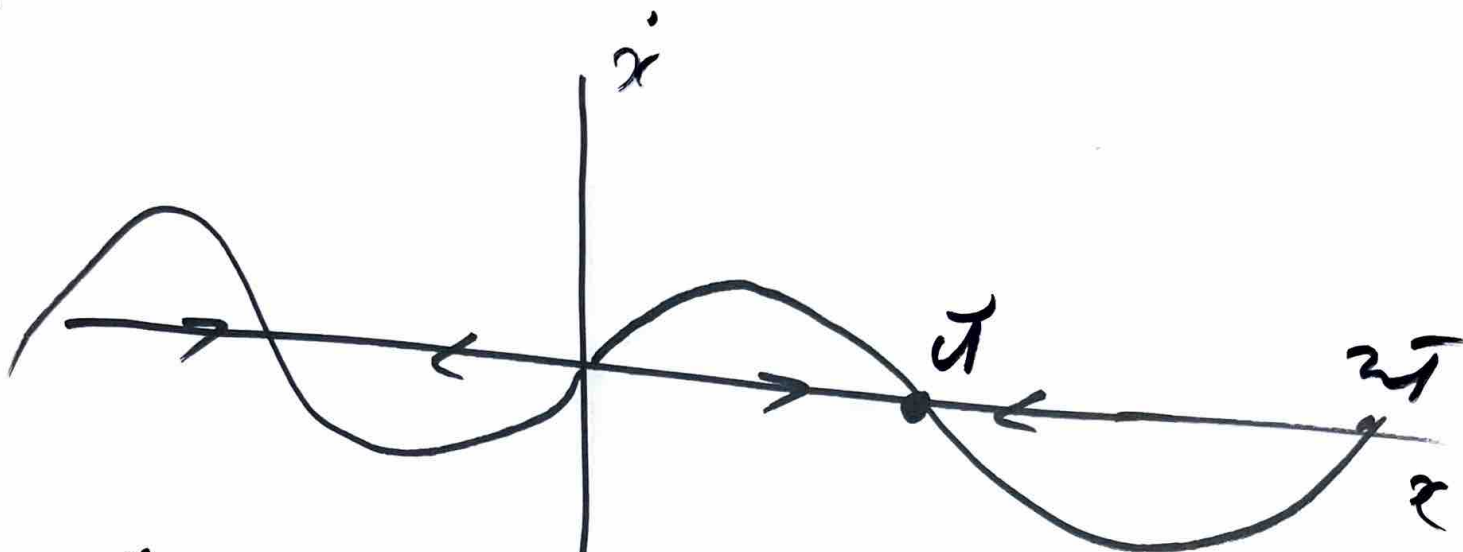
-10-

$$\bullet x(t=0) = 0 \Rightarrow x(t) \equiv 0$$

$$\bullet x(t=0) = 1, \quad x(t) \equiv 1$$

$$\bullet x(t=0) = m1, \quad x(t) \equiv 1 \cdot m$$

$$m = 0, 1, 2, \dots$$

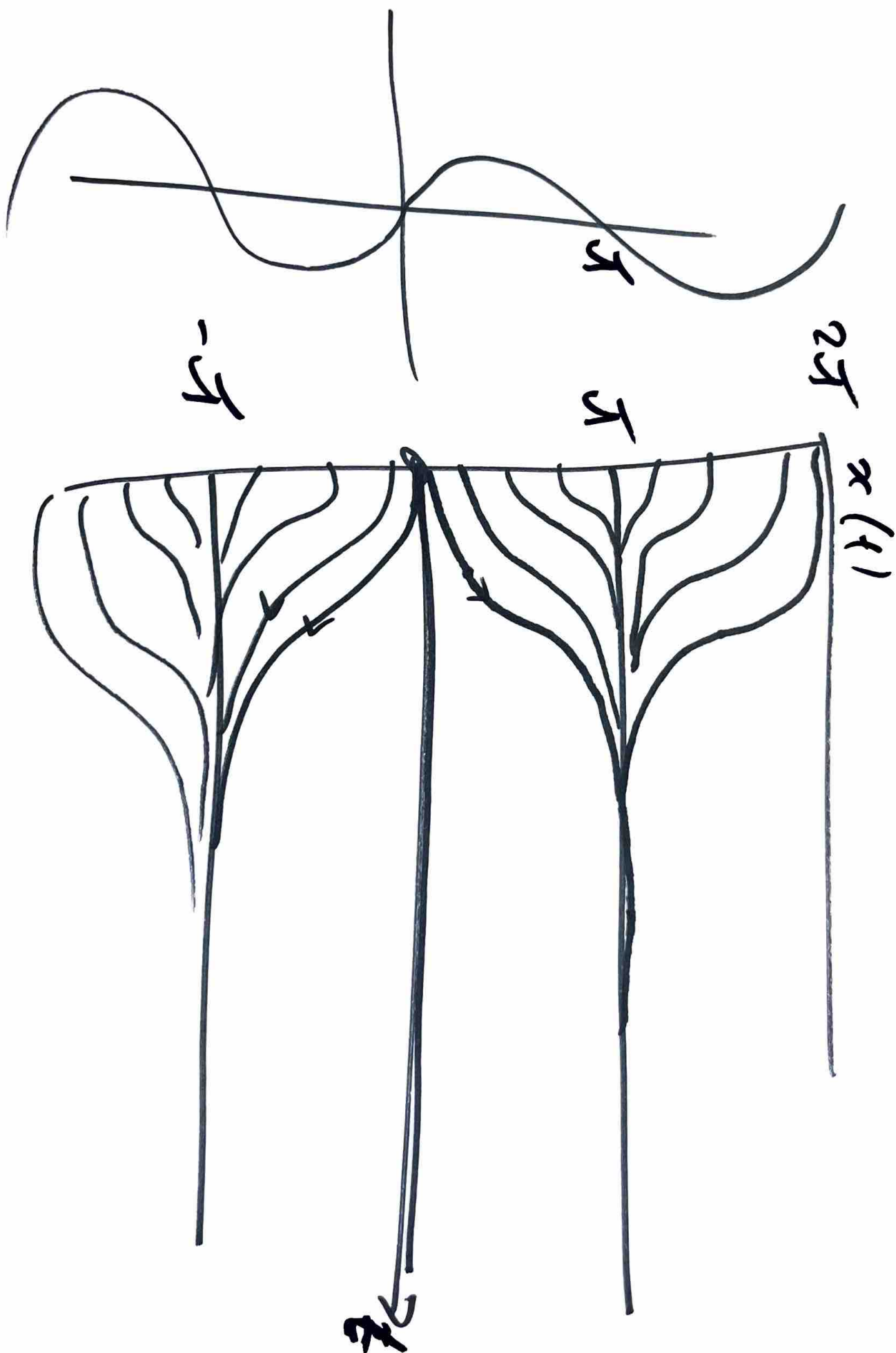


x^* is a fixed point

iff $f(x^*) = 0$

For this example $x^* = m\pi$; $m = 0, \pm 1, \pm 2, \dots$

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-13-

To solve $\dot{x}(t) = f(x(t))$

① Find all fixed points:

Find x^*

such that $f(x^*) = 0$

② Analyze regions between fixed points:

IF $f(x) > 0 \Rightarrow$ flow to the right

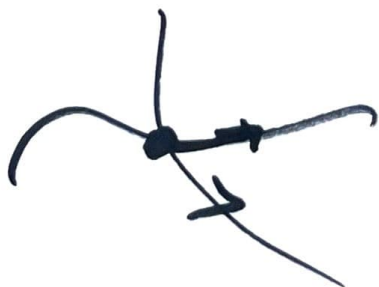
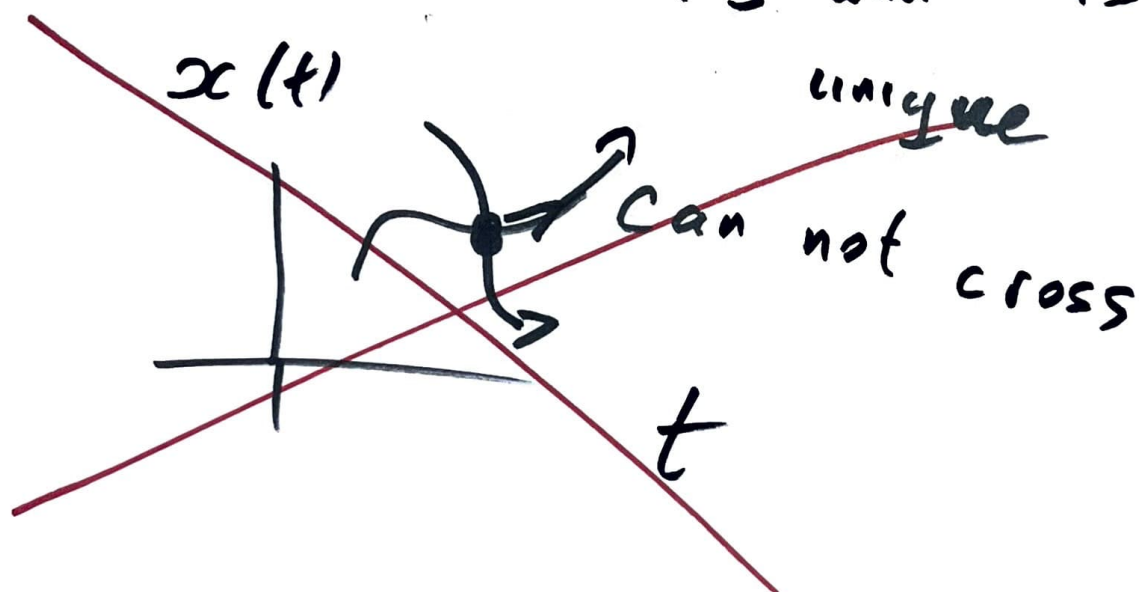
— " — $f(x) < 0$ — " — " — left

— " — $f(x) = 0$ stay where you are

-14-

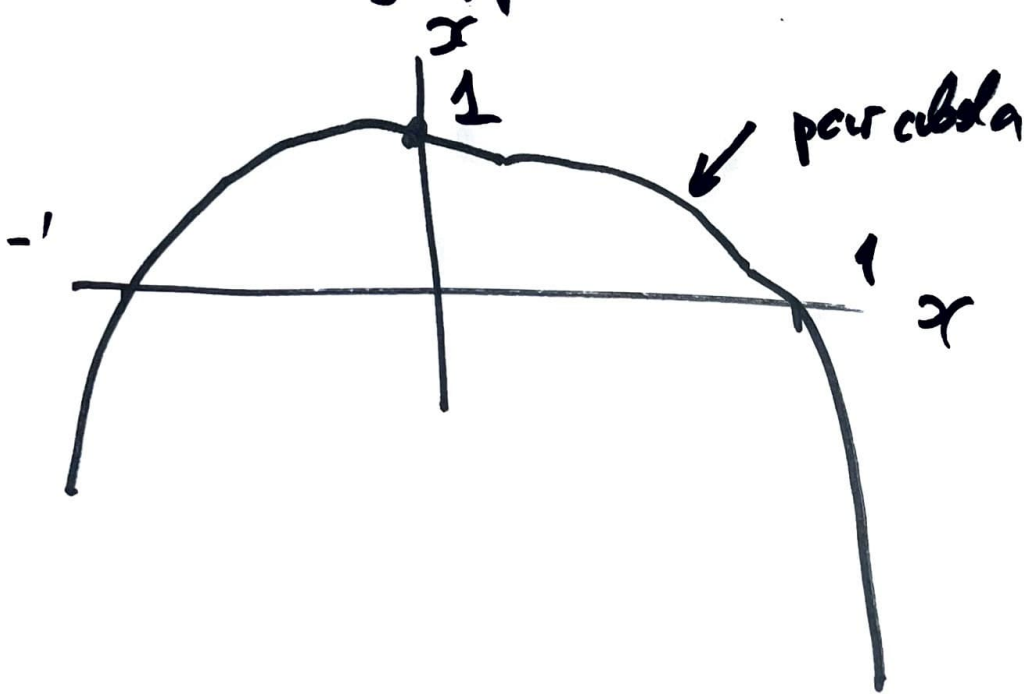
Existence and uniqueness
if $f(x)$ is C^1 , continuously
differentiable

$\Rightarrow x(t)$ exists and is

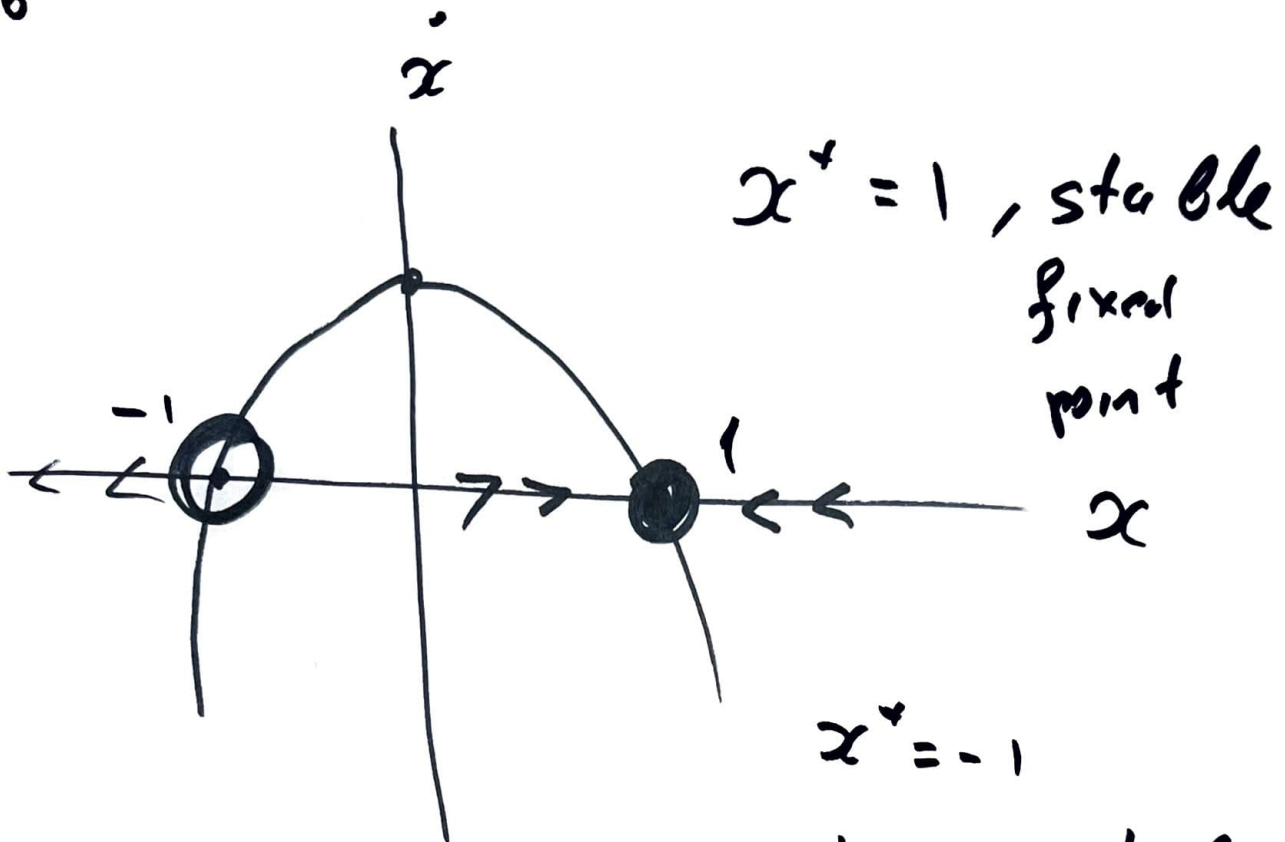


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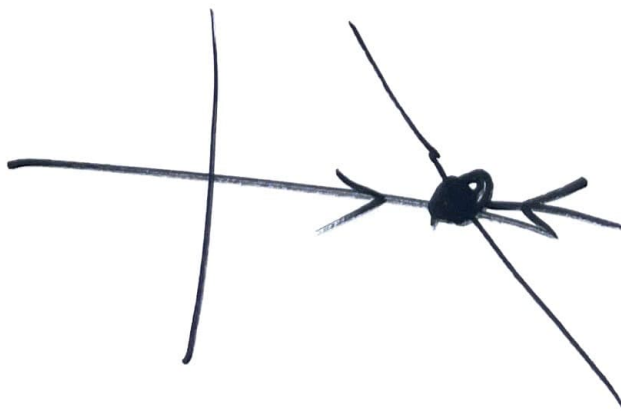
Study $\dot{x}(t) = 1 - x^2(t)$
find fixed points analyze
stability, plot $x(t)$



-16-



$x^* = -1$
is unstable fixed point



-17-

$$\dot{x} = 1 - x^2 = \frac{dx}{dt}$$

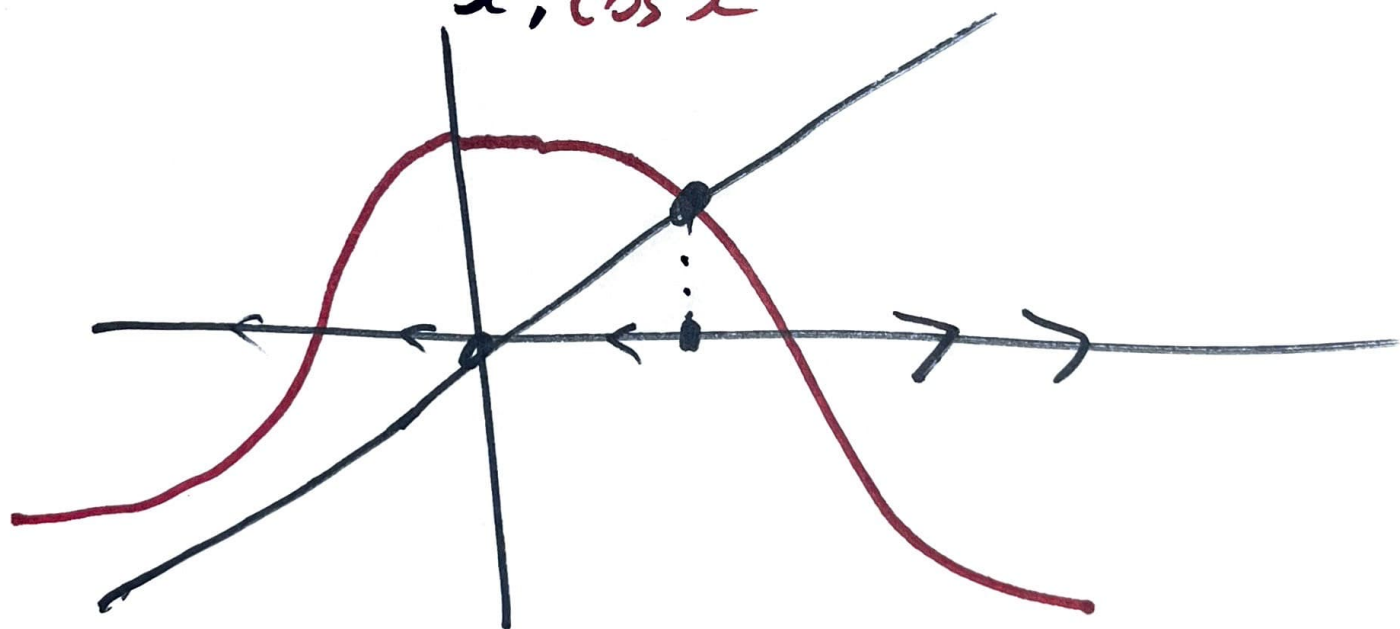
$$\begin{aligned}\int \frac{dx}{1-x^2} &= \text{Arc Tan } x = t + c \\ x &= \tan(t + c)\end{aligned}$$

18 Example [same steps]

$$\dot{x} = x - \cos x$$

Find fixed points, analyze stability,
find trajectories.

$x, \cos x$



$x^* = \cos x^*$ - does not have
analytical
 x^* is crossing of $x, \cos x$ set